

Remote Sensing of Correlation, Seasonality, and Trends of Urban Heat Island and Urban Pollution Island: A Comparative Study of Four Major U.S. Cities

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Background

Urban Heat Islands (UHI) and **Urban Pollution Islands (UPI)** are both byproducts of rapid urbanization and the manifestations of intensive modification of the built environment. UHI refers to the urban phenomenon where temperature tends to be higher in urban zones than surrounding non-urban zones due to modification of energy balance, whereas UPI refers to the spatial and temporal variations in pollutant concentration attributable to anthropogenic urban features and human activities (Ulpiani, 2021). While UHI and UPI are well-documented urban problems, they are often analyzed separately despite evidence of a strong correlation between their spatial configuration with a coefficient of 0.62, as demonstrated by a study in Berlin due to the complexity of their linkage (Li et al., 2018). Moreover, their relationship is not rigorously studied in the context of U.S. cities whose populations could potentially experience severe health impacts due to their compounding effects.

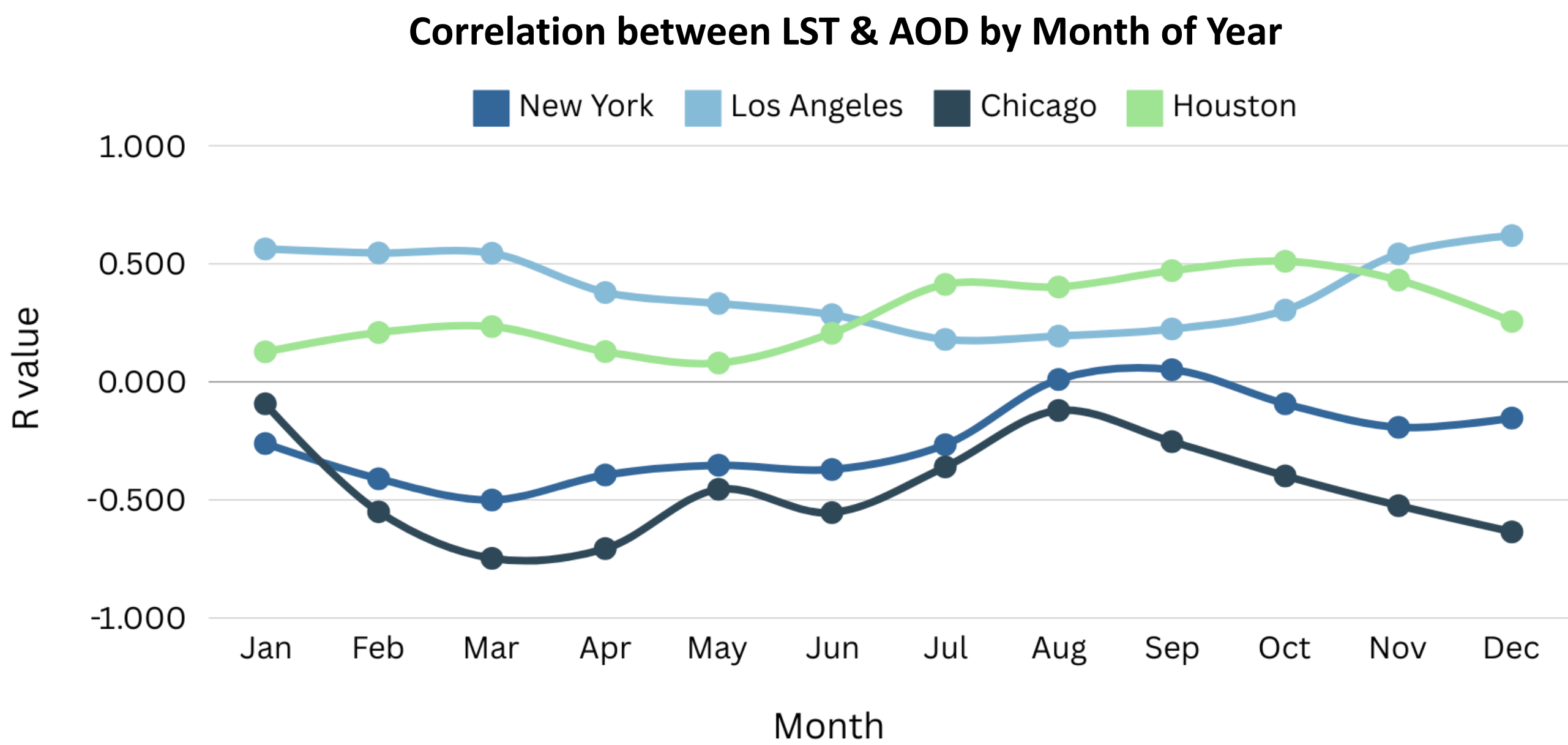
Research Goal

- 1) Analyze the seasonal correlation between UHI and UPI
- 2) Understand the seasonal dynamics of LST and AOD
- 3) Identify the overall trend of UHI and UPI

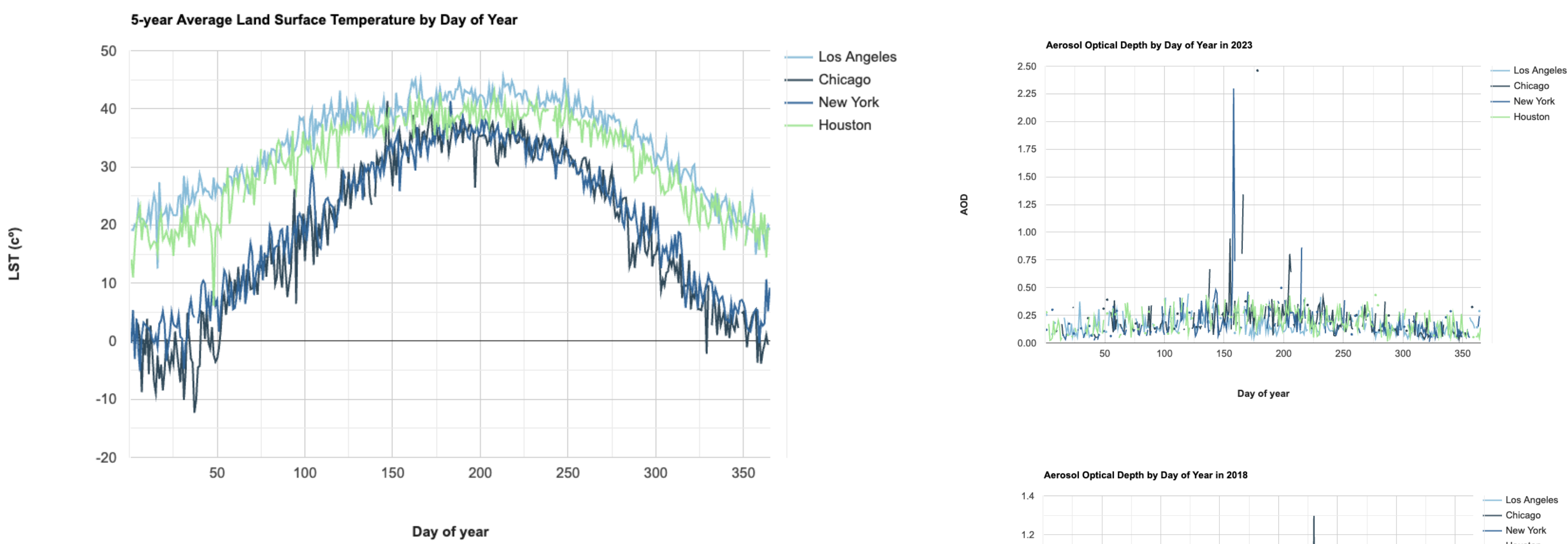
Methods & Study Area



Correlation & Seasonality



There is limited variation within a year across all study areas where Los Angeles and Houston in general have a positive relationship while Chicago and New York have a negative relationship. A vast majority of the correlation results are considered statistically significant ($p < 0.05$) except for August and September in New York and May in Houston.

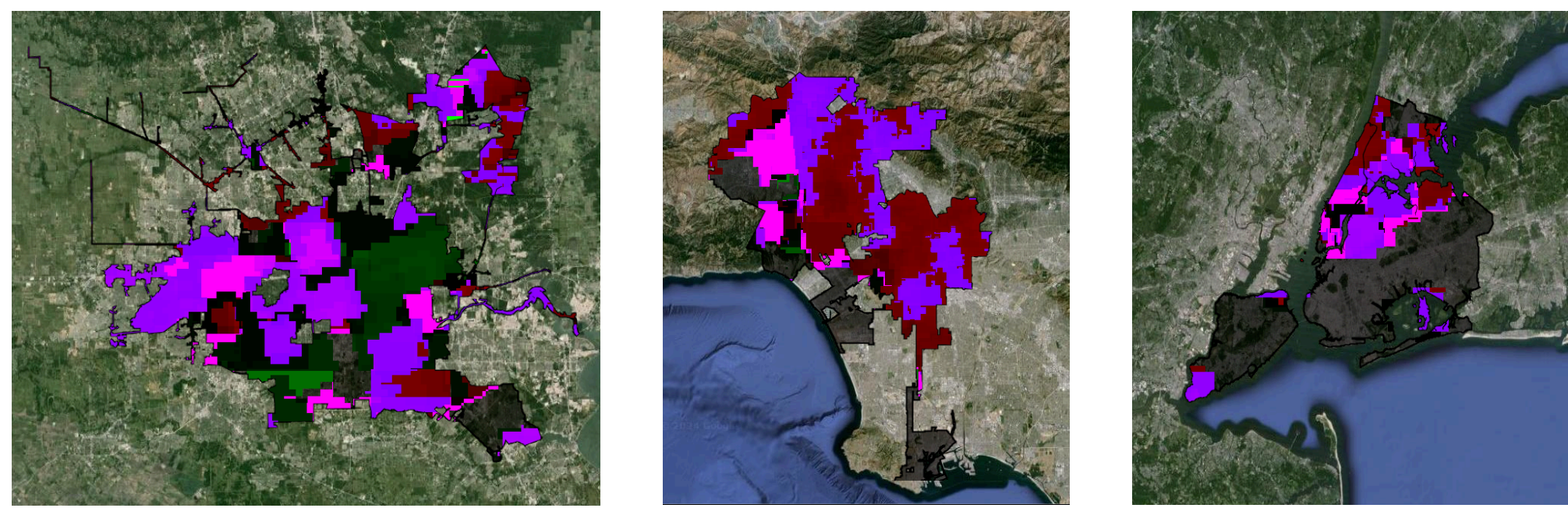
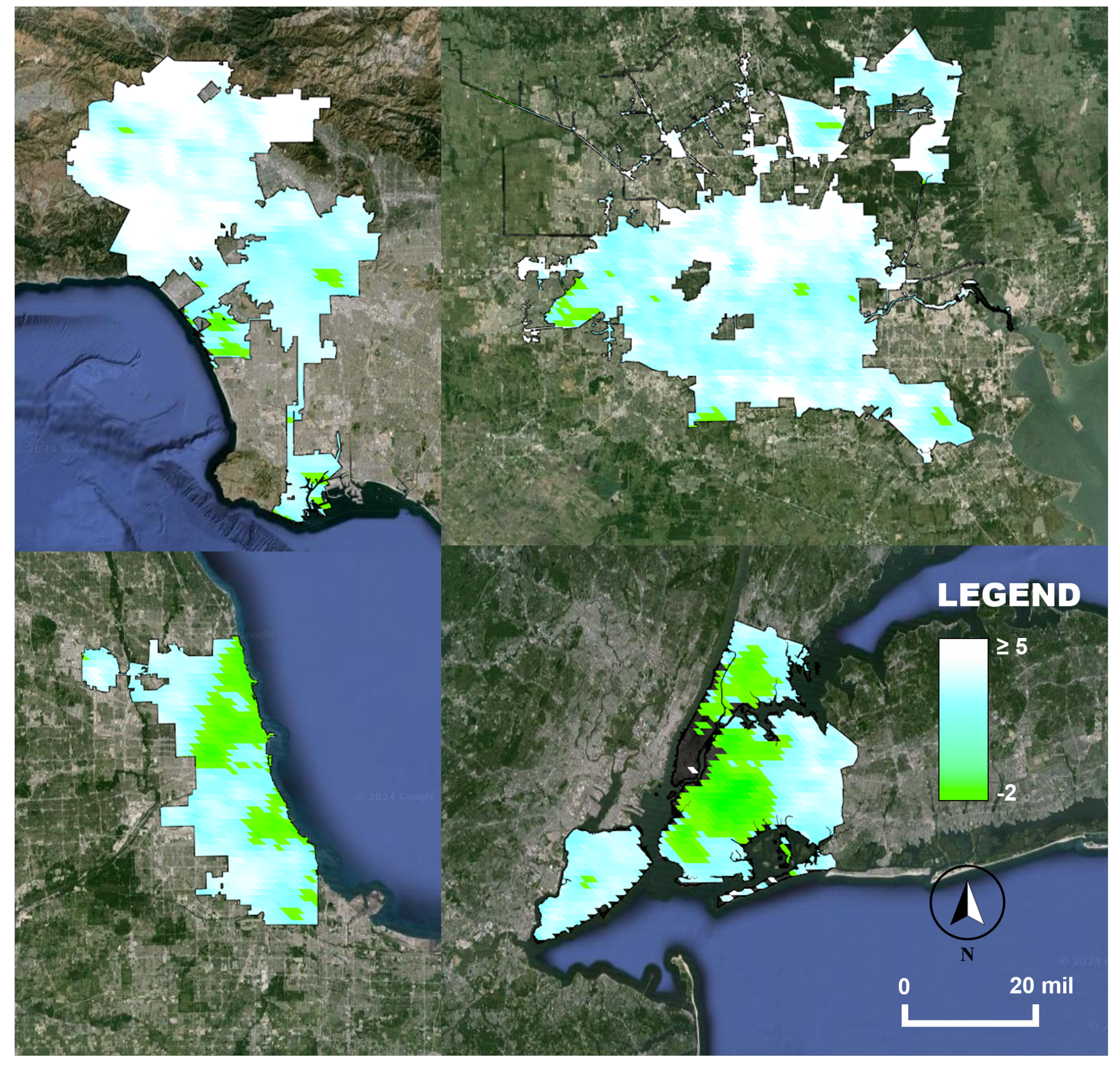


Compared to AOD, LST has a much more distinct seasonality where temperatures start increasing at the beginning of the year and peak between days 150 and 200 before declining. Conversely, AOD values do not manifest seasonal patterns in both 2018 and 2023. Values fluctuate between 0 and 5 for the majority of the year. Moreover, there is no significant difference in terms of seasonality across all study regions, suggesting consistent temporality of UHI and UPI impact.

Conclusion & Limitation

It is clear that spatial correlations exist between UHI and UPI and the relationship differs from city to city. Cities with a negative relationship are located in the northern part of the continent while cities with a positive relationship are located in the southern part where they experience higher heat stress due to higher temperatures. However, a bigger pool of study areas is needed to validate this theory. The trend analysis not only insinuates the aggravation of UHI and UPI ascribable to land cover modification and climate change but also illustrates the complex interaction between UHI and UPI. While understanding their dynamic co-existence is vital, it is also crucial to evaluate their impact on public health to inform healthy city strategies.

Trend Analysis



All study cities illustrate a steady increase in terms of LST over the 5-year period, and Los Angeles and Houston have the highest scale values, which are 3.5 and 3.1 respectively. Similarly, all regions manifest a positive trend in terms of AOD. The result of Chicago is disregarded due to the number of masked pixels in the region caused by the sensitivity to cloud contamination.

References

Li, Huidong, et al. "Interaction between urban heat island and urban pollution island during summer in Berlin." *Science of the total environment* 636 (2018): 818-828.
Ulpiani, Giulia. "On the linkage between urban heat island and urban pollution island: Three-decade literature review towards a conceptual framework." *Science of the total environment* 751 (2021): 141727.